AI ACADEMY · LEVEL 2 · GRADE 7 · WEEK 8 · STUDENT LESSON

Operational Labels

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

Define the label. Check the boundary. Investigate disagreement.

Use an ordered calendar guide and a separate book-length policy to explain unknowns, equality and agreement.

Keep one habit: Observable criteria → individual labels → source-backed comparison → versioned repair.

MAIN LESSON · SECTION 1

How can two people apply the same label reliably?

> Replace vague impressions with observable criteria and an explicit boundary.

Your learning goals

I can apply a **category** definition using explicit **evidence** and boundary rules.

I can distinguish a known failure from information that is missing or unreadable.

I can use **disagreement** to improve a **label**ling guide without mistaking **agreement** for truth.

Recall the difference between missing and known

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

A transcript leaves out an unclear word. Why should a later system not simply treat that missing word as a known value?

This week’s deciding skill

By the end, I can resolve **label disagreement** by checking an **operational definition** at equality.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Your learning target

Remember before reading

Without opening the old lesson, explain one key idea from Rows Columns and Data Types. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Loops and Stopping. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Decompose a Goal. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 8: Operational **Label**s

A **label** is reliable only when different people can apply its definition consistently.
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THINKER MISSION

Write and test definitions for ‘easy book’ and ‘hard book.’

Thinker name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

This week’s end goal

MAIN LESSON · SECTION 2

Start with a vague instruction: good cards

> A label needs a shared meaning for a particular task.

Two reasonable interpretations

Two students are sorting fictional event cards for a classroom calendar. Their first instruction is Put the good cards together. Mina calls a colourful card good. Theo calls a plain card good because its room is easy to read.

They disagree even though both followed a reasonable interpretation of the word good.

Define what can actually be checked

A **category** is useful for a task when people can tell what **evidence** makes an example belong to it. An **operational definition** states the **observation**s or checks to perform.

For this paper exercise, we will check two printed fields: an event day and a room code. The task is to check the record against a classroom policy, not to judge the quality of the event or the person who wrote it.

Why label quality matters to later models

Labels depend on the definition in use. A model fitted to **label**s can learn the distinctions those **label**s encode, including mistakes and inconsistencies.

Before asking a model to copy a **label**, ask whether people have a clear, useful guide for assigning it.

▼ Earlier wording, alternatives and picture transcripts

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



MYSTERY

The learners must use operational **label**s to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

MAIN LESSON · SECTION 3

Apply the calendar guide in its stated order

> Known failure takes priority over unknown information in this policy.

Read the field contract and three labels

Our fictional calendar accepts a whole-number day from 1 through 30, including both endpoints, and exactly one room code, A or B. The month is already fixed for this exercise. A blank or unreadable field counts as unknown.

The categories are revise, review and ready. Use the checks in the order shown.

Follow the ordered checks

Three ordered checks for a calendar card

CHECK IN THIS ORDER	EVIDENCE	RECORDED LABEL
First check known values	Any known field fails its criterion	Revise even if another field is unknown
Then check missing information	No known failure but a field is blank or unreadable	Review because evidence is incomplete
Otherwise all checks have passed	Day is 1 through 30 and room is A or B	Ready under this record policy

Stop at the first matching row. The order resolves a card with both a known failure and an unknown field. Ready describes the printed record under this policy; it does not confirm that the event exists or has permission to run.

Contrast unknown and known failure

Known failure and unknown are different. Day 31 is known and outside the accepted range, so the card needs revision. An unreadable day might be valid or invalid; without another known failure, it needs review.

Never silently turn a blank into zero just to make a comparison possible.

Point to a deciding step

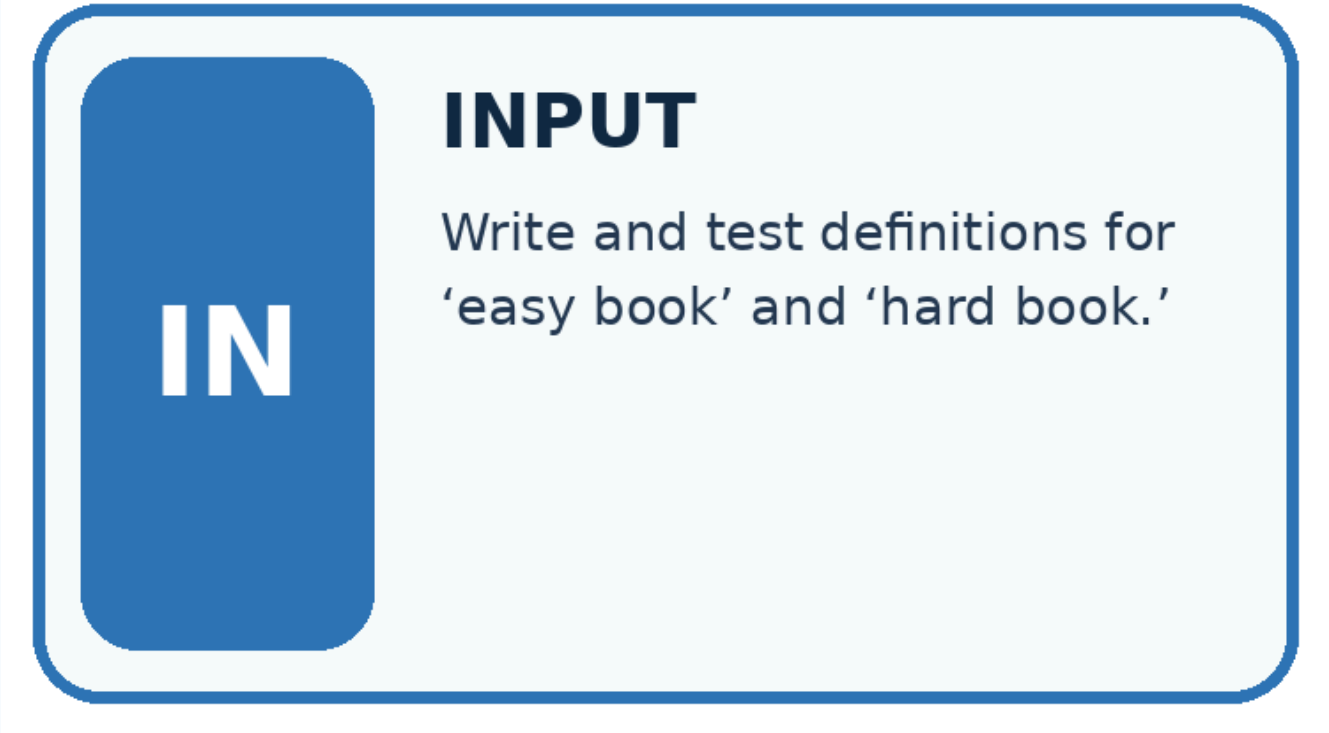
- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ **Earlier wording, alternatives and picture transcripts**

VISUAL **MECHANISM**

Input → **Process** → **Output**

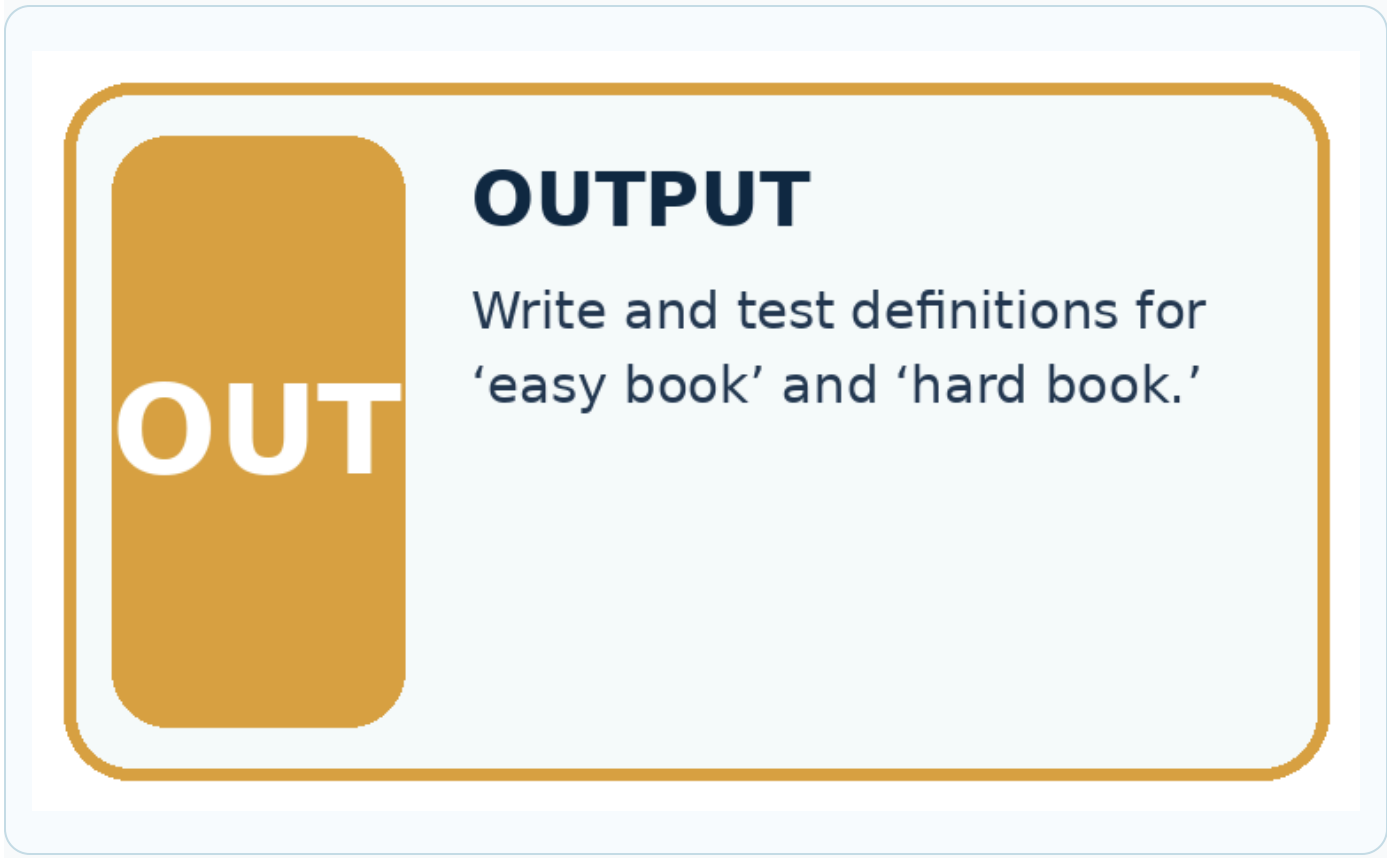
Trace the information. At each stage, ask what changed and how you could test it.



↓


PROCESS
A label is reliable only when different people can apply its definition consistently.

↓


OUT
OUTPUT
Write and test definitions for ‘easy book’ and ‘hard book.’

Which stage could fail even when the other two are correct? Explain:

> **Write or draw your reasoning and evidence.**

A closer explanation

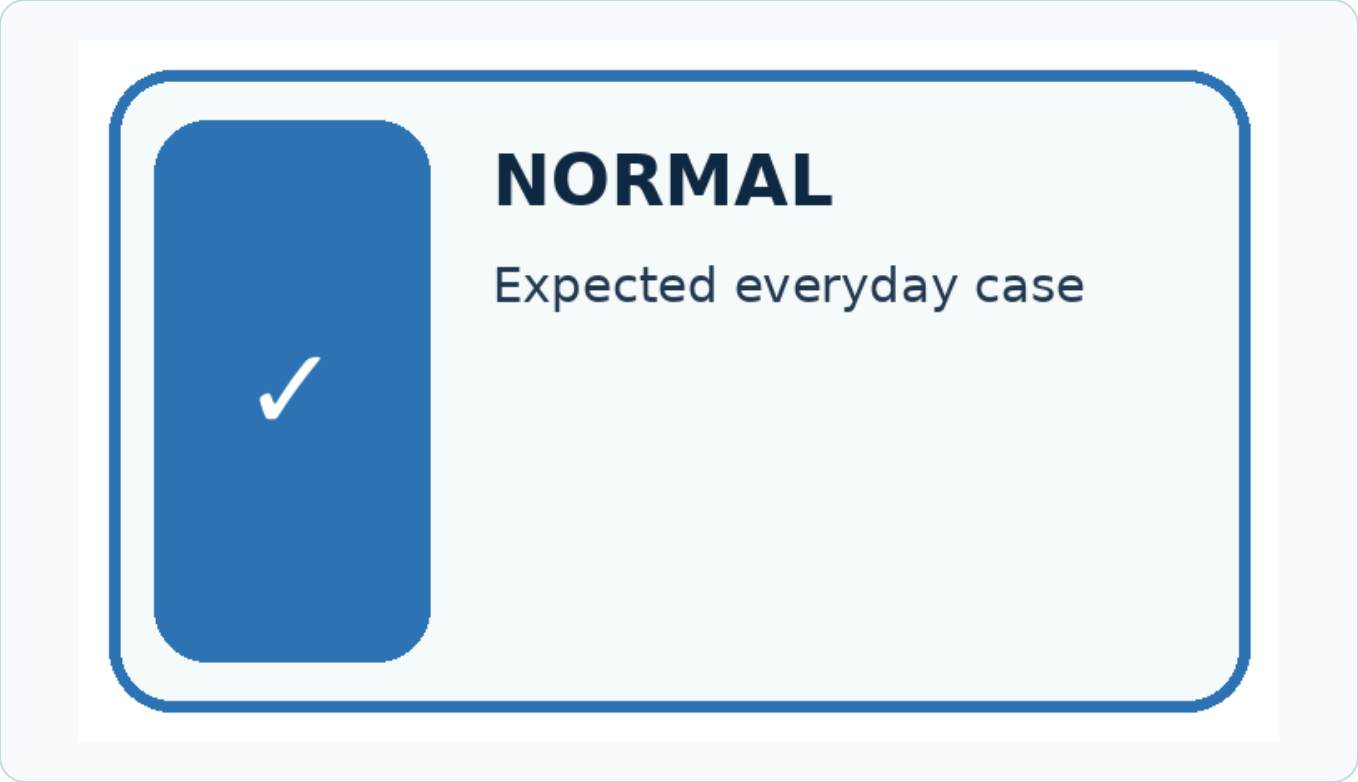
Connect the picture and the explanation

MAIN LESSON · SECTION 4

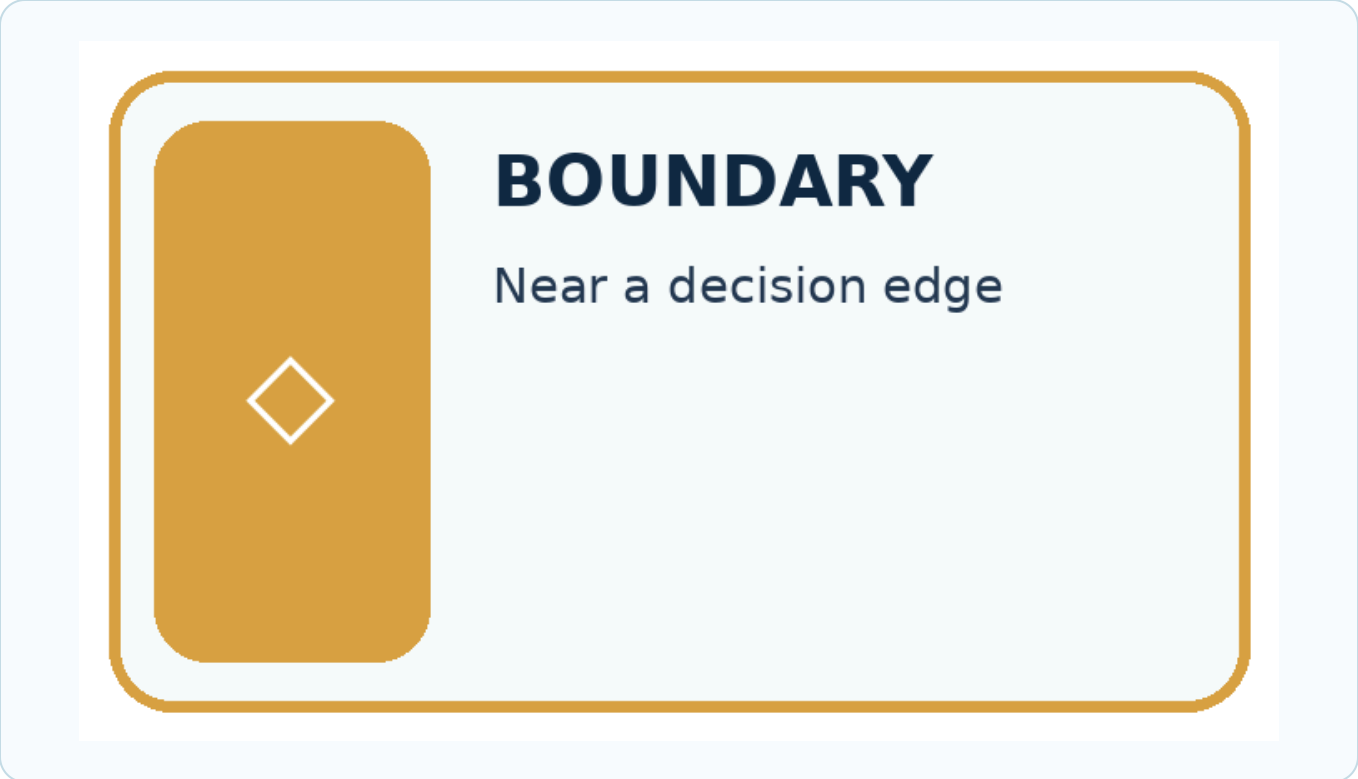
Compare a pass, a boundary and incomplete evidence

> Use the guide rather than confidence or appearance.

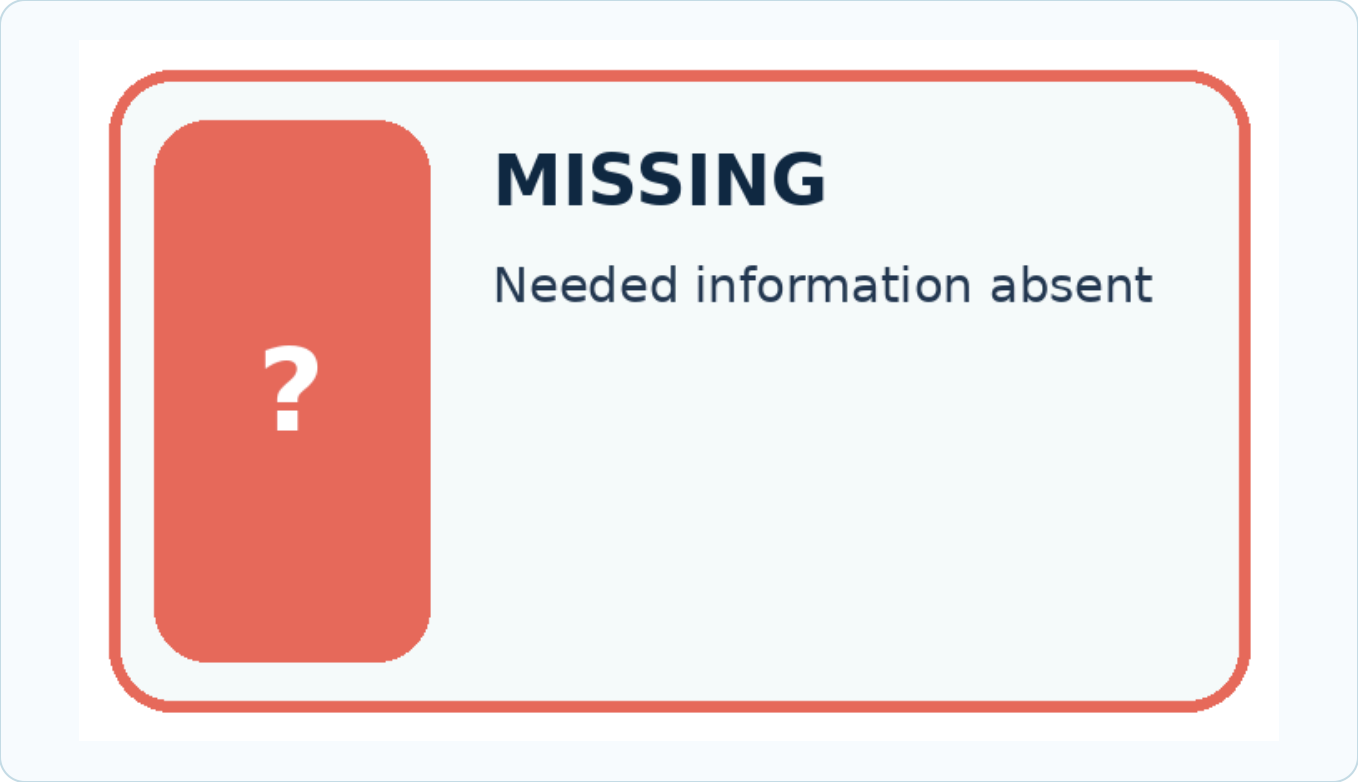
Four pictures with a purpose



Normal case: predict the result and deciding evidence.



Boundary case: identify the exact limit.



Missing information: name what must be clarified.



Transfer case: explain which procedure still applies.

Choose a revealing case

- Which case is most likely to reveal a hidden assumption? Why?

> **Write or draw your reasoning and evidence.**

Predict the supplied changed case

- Change: Change the documented policy to fewer than 80 instead.

▼ Check after predicting

Prediction: A's original **label**s now match; B must change its equality behavior.

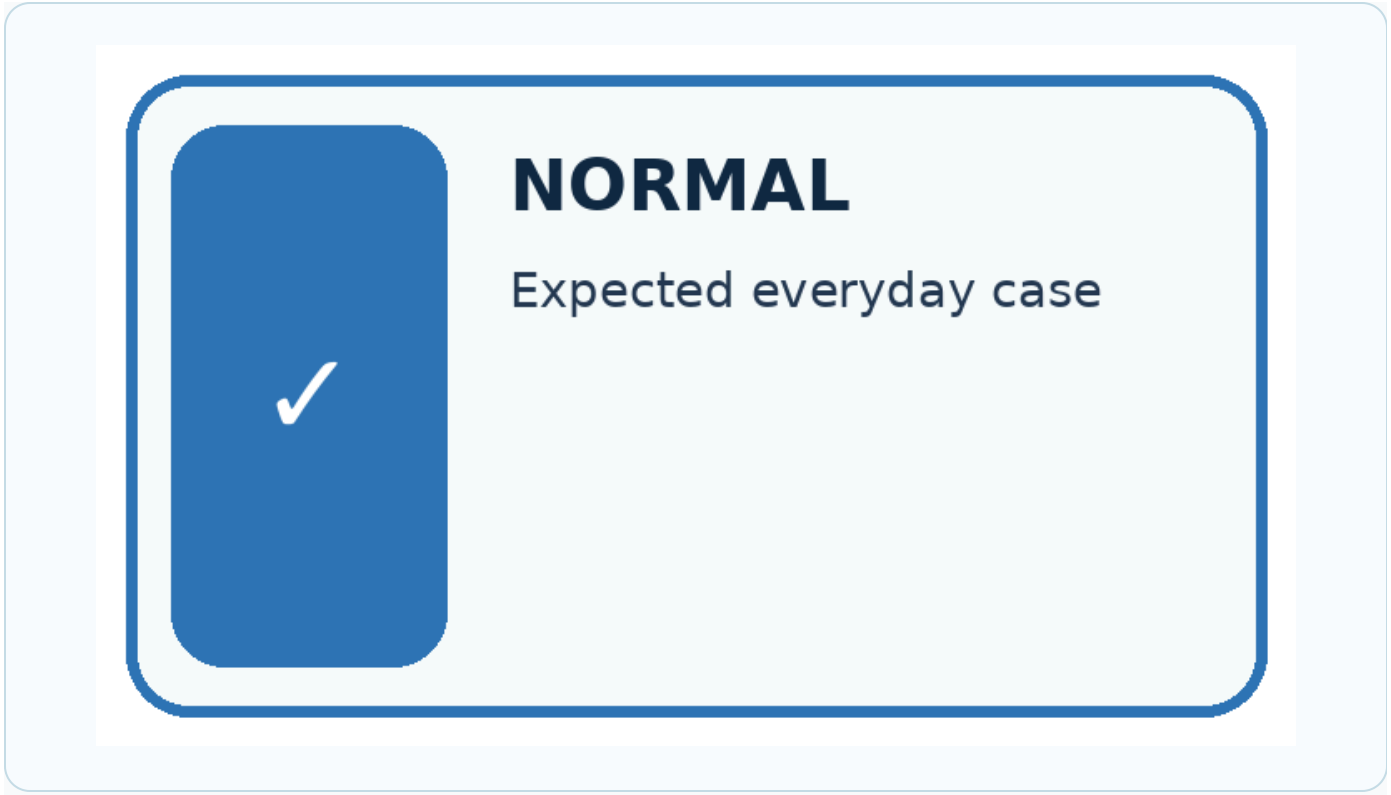
Reason: The authoritative definition changed, not the physical books.

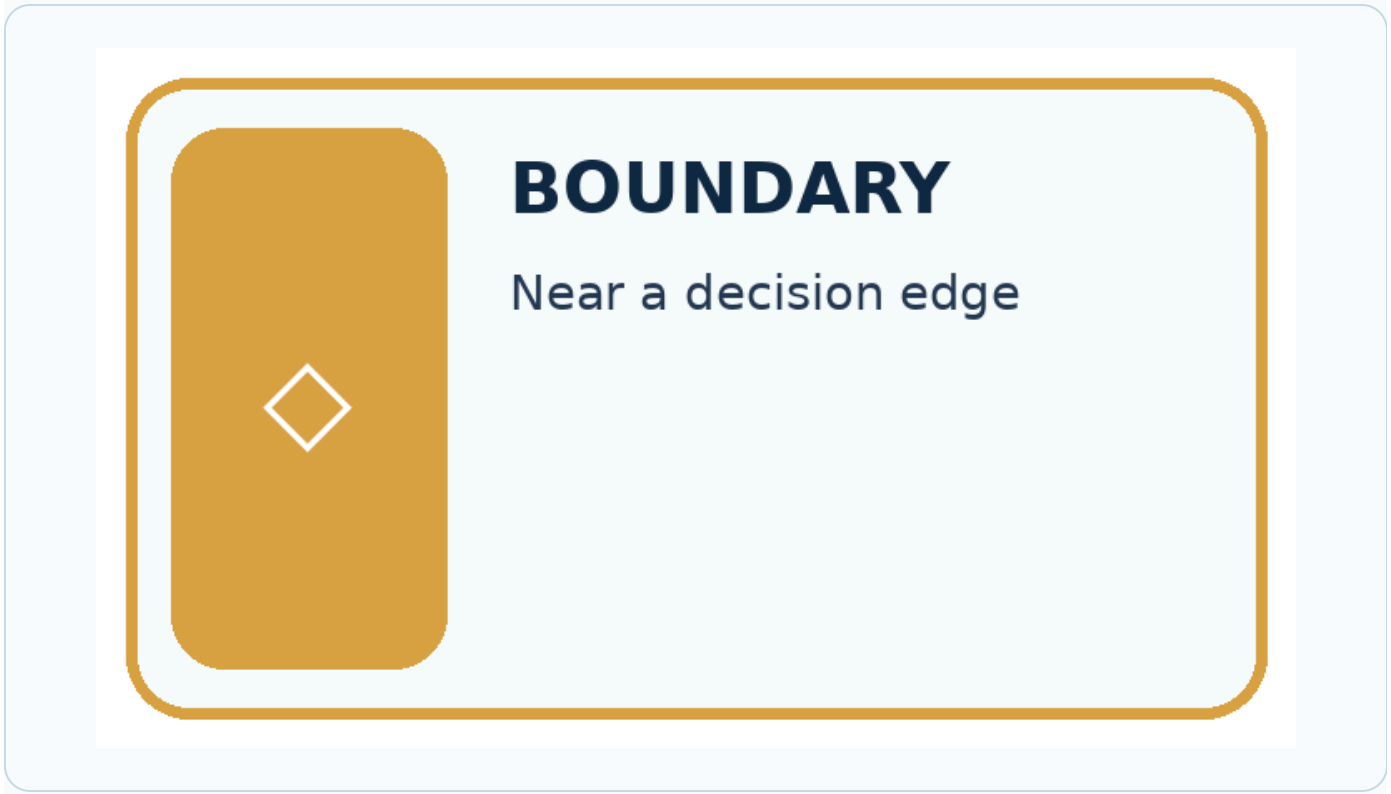
▼ Earlier wording, alternatives and picture transcripts

CASE GALLERY

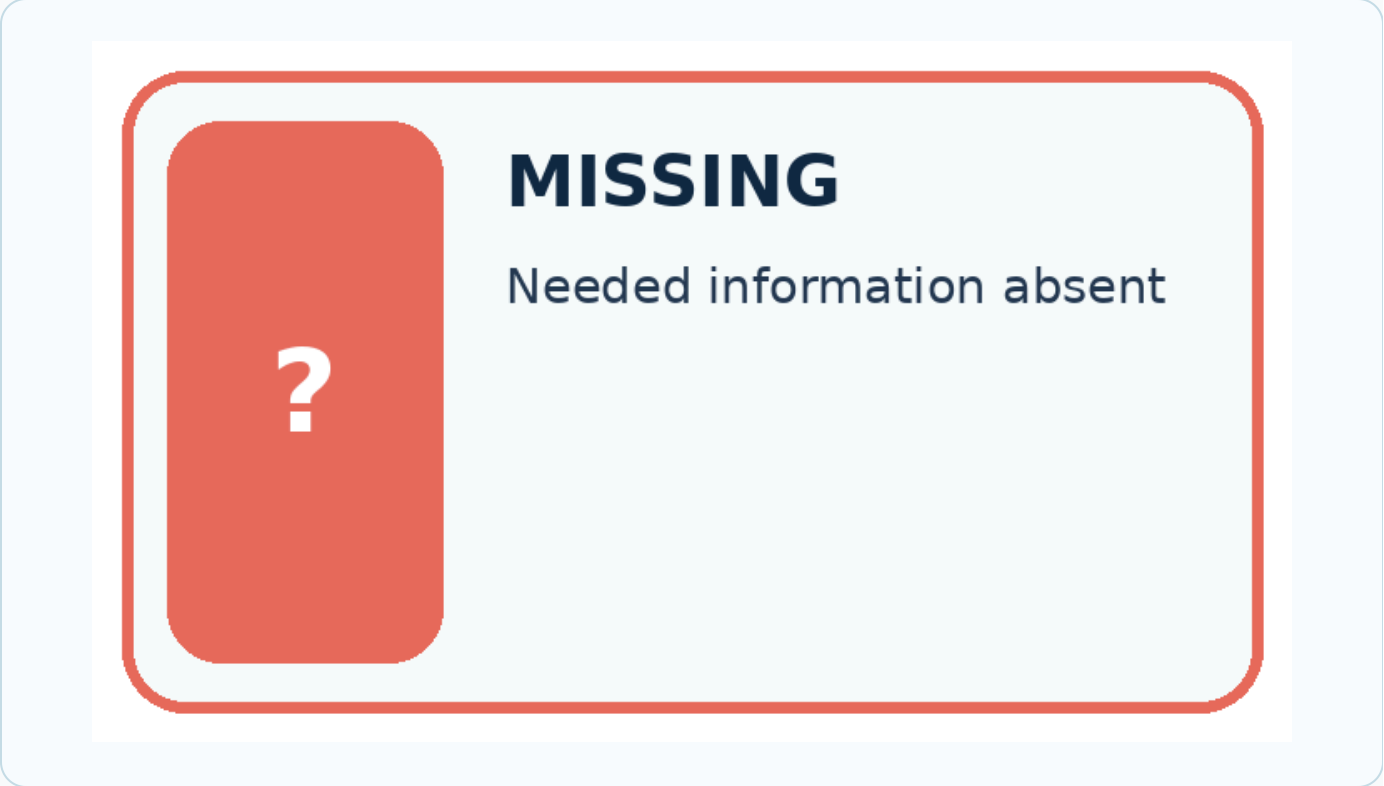
Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.





NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Trace calendar labels, then a separate page-count boundary

> Keep the event policy distinct from the compact-book policy.

Trace the four calendar records

Card fields	Relevant evidence	Label and reason
Day 12 room A	Both values satisfy the criteria	Ready because both checks pass
Day 31 room A	31 is outside 1 through 30	Revise because a known value fails
Day 12 room unreadable	No known failure but the room is unknown	Review until the room is checked
Day 31 room unreadable	The day already fails although the room is unknown	Revise because the first check has priority

Explain the mixed failure/unknown case

For the last card, first read only what is known: day 31. Compare it with the allowed range and record a failure. The guide therefore selects revise before the missing-room check.

Checking the room later could reveal another problem, but it cannot make day 31 satisfy this policy.

A clear explanation cites the particular criterion. Saying It looks wrong is not enough. Write Day 31 exceeds the accepted maximum of 30 instead.

That explanation lets another person reproduce the check and challenge it if the recorded value or policy is different.

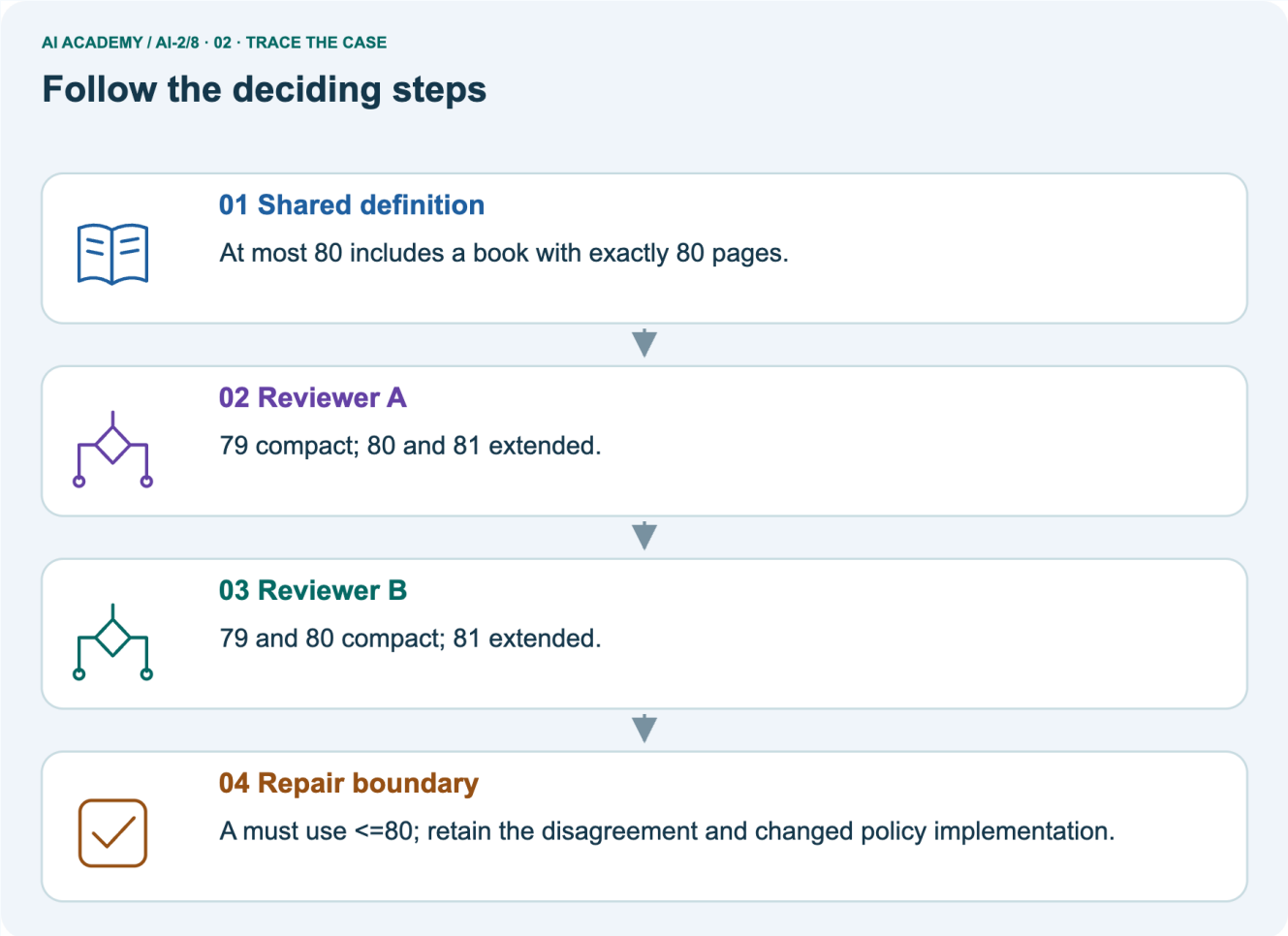
Introduce the separate compact-book policy

Two fictional catalogue reviewers **label** books compact or extended.

Agreed policy: compact means at most 80 pages; extended means more than 80.

Reviewer A uses pages<80; reviewer B uses pages<=80. Inspect books of 79,80,81 pages.

Follow the reviewers' different equality checks



Follow the labelled picture with the matching explanation. Open the vector diagram

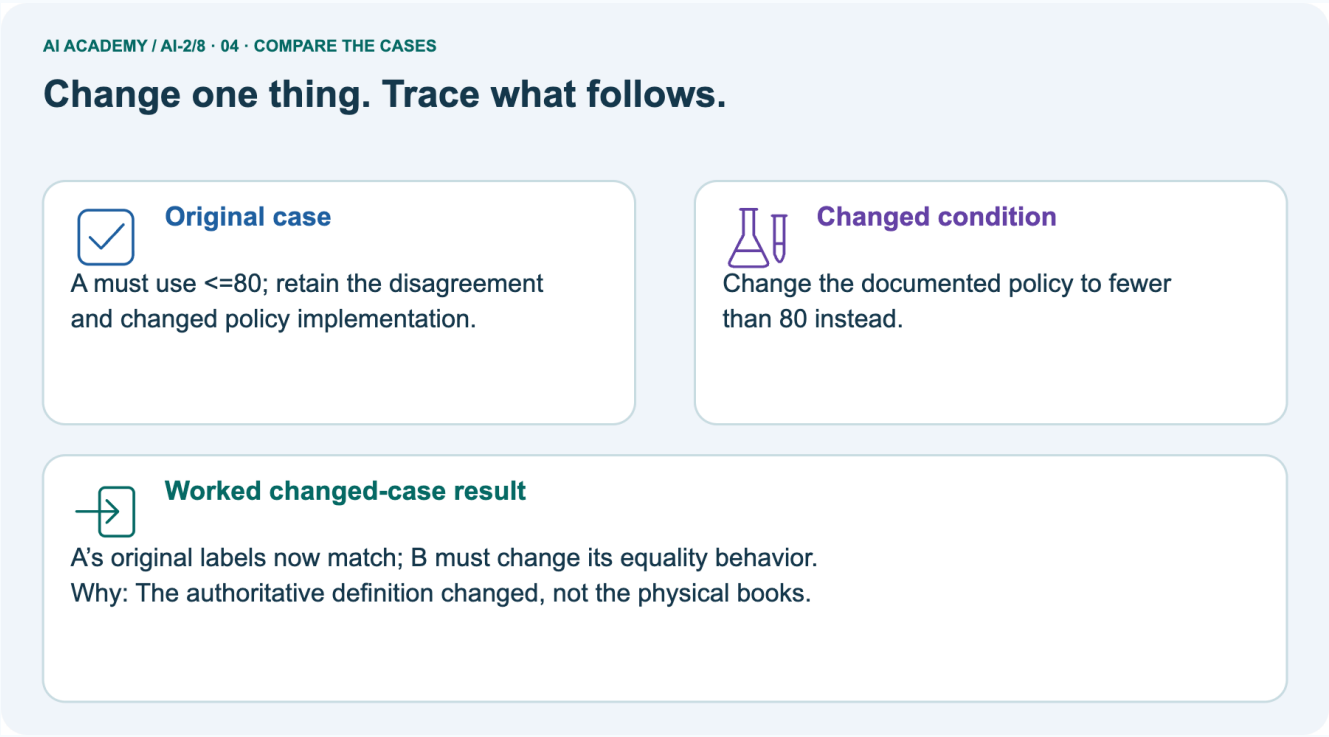
The reviewers agree on two of three examples, but that fraction does not decide which definition to keep. The documented policy decides the equality case. Repairing A's implementation aligns its **label**s with the agreed meaning.

Rechecking 79 and 81 ensures the boundary fix preserves adjacent decisions.

Predict a changed documented definition

- Change one thing: Change the documented policy to fewer than 80 instead. Predict the result before checking the explanation. What remains the same?

▼ Check only after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain disagreement and the category's limits

- Read the **label**led picture: Why can we not resolve the **disagreement** merely by taking a majority of reviewers?

- A shared definition improves consistency but does not prove the **category** captures every learner's needs.

▼ Earlier wording, alternatives and picture transcripts

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

Worked evidence for Operational Labels

STEP	EVIDENCE AND REASONING
1	Weak label: ‘easy book.’
2	Operational rule: easy for this reader if difficulty level ≤ 2 and unfamiliar words per page ≤ 3.
3	Two labelers test the same 8 books.
4	Agreement of 7/8 identifies one disagreement to review. Check the source, each annotator’s application and the label rule before choosing a repair.

Trace these corrected steps. Keep the stated metric, assumptions and interpretation when you change the case.

1. Weak **label**: ‘easy book.’

2. Operational rule: easy for this reader if difficulty level ≤ 2 and unfamiliar words per page ≤ 3.

3. Two **label**ers test the same 8 books.

4. **Agreement** of 7/8 identifies one **disagreement** to review. Check the source, each **annotator**'s application and the **label** rule before choosing a repair.

Recalculate, retrace, or restate the reasoning in your own representation:

> **Write or draw your reasoning and evidence.**

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Operational Labels

The facts we will use

AI ACADEMY / AI-2/8

Operational label boundary

	Pages	A labels	B labels
Book 1	79	compact	compact
Book 2	80	extended	compact
Book 3	81	extended	extended

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/8

Operational Labels

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Shared definition

At most 80 includes a book with exactly 80 pages.



2. Reviewer A

79 compact; 80 and 81 extended.



3. Reviewer B

79 and 80 compact; 81 extended.



4. Repair boundary

A must use <=80; retain the disagreement and changed policy implementation.

CHANGE ONE THING

Change the documented policy to fewer than 80 instead.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Shared definition	At most 80 includes a book with exactly 80 pages.
2. Reviewer A	79 compact; 80 and 81 extended.
3. Reviewer B	79 and 80 compact; 81 extended.
4. Repair boundary	A must use <=80; retain the disagreement and changed policy implementation.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Agreement counts matches; it does not establish correctness

> Compare label reasons with the policy and source facts.

What four matches out of six tells us

If two **annotator**s agree on four of six cards, we know that four assigned **label**s matched. We do not yet know whether those **label**s followed the guide. Both **annotator**s might have made the same mistake.

Check their reasoning against the criteria and, where necessary, the source record.

Keep reality and policy versions separate

Even perfect **agreement** under our calendar policy would not prove that the room is really booked. It would only show **agreement** about whether the printed fields pass the specified checks.

The policy itself may need changes for a real calendar. Record the policy version so examples **label**led under different instructions are not mixed without review.

Challenge perfect agreement

> **Two people label every card ready. Explain why their agreement alone does not establish correct labels or confirm the events**

▼ **Earlier wording, alternatives and picture transcripts**

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: People who mean well will always create identical labels. Strong move: locate evidence, change one cause, and retest.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Practise the deciding criterion before comparing labels

> Record individual answers before discussion.

Resolve the inclusive day-30 boundary

Mina **label**s day 30, room B as ready; Theo **label**s it revise because he thought the range stopped before 30. The **disagreement** points to a boundary interpretation. The guide says including both endpoints, so 30 passes.

The appropriate repair is to clarify and practise the inclusive boundary, not to change the **label** merely because one student is more confident.

Try known failure versus unknown

> **Label day 0 with room A and day 18 with a blank room. Cite the exact rule for each and explain why the two cards need different next steps**

Keep original individual labels

Ask **annotator**s to **label** examples independently before comparing. If they copy one another, **agreement** no longer tells us whether the guide can be applied independently.

Keep the original **label**s and reasons when discussing differences, then record any corrected **label**s separately.

Replace vague instructions in W2

Use workbook W2 for the connected example “Categories and operational **label**s”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Repair a missing-value misconception

- Use the worked case for Categories and operational **label**s. Proposed explanation: “A blank is the same as a failing value”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the separate book-policy boundary

Why can we not resolve the **disagreement** merely by taking a majority of reviewers?

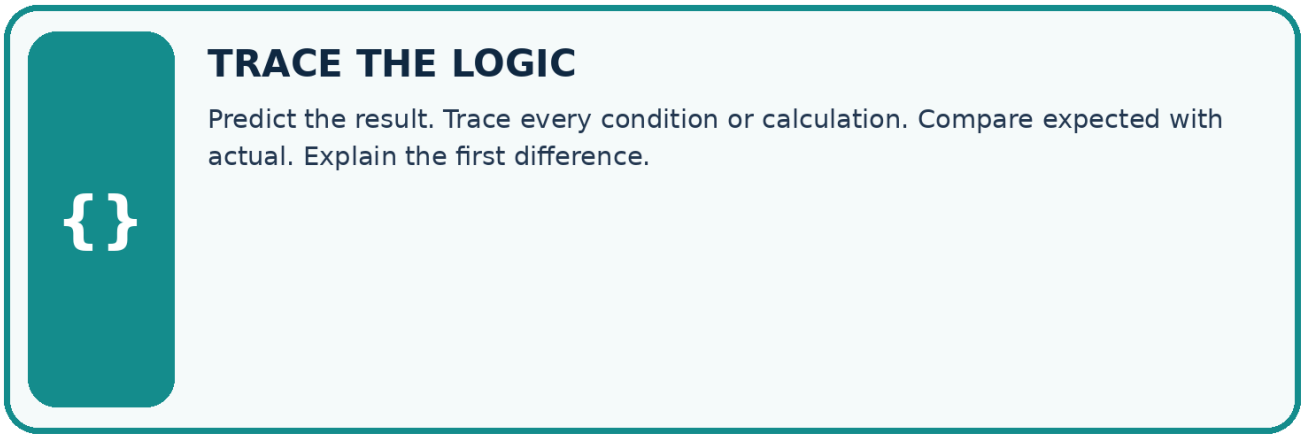
This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Earlier wording, alternatives and picture transcripts

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



SET **evidence**_record PREDICT expected_result RUN one normal case and one boundary case COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST EXPLAIN result, limitation, and next safe step

Trace one **input**. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Practise with less help: Replace a vague instruction

Check your reasoning

Explain the picture

My explanation using a pictured step and the given facts

MAIN LESSON · SECTION 9

Label six supplied cards independently, then audit disagreements

> Separate agreement, correctness and policy changes.

The guide used by both annotators

Use calendar guide version 1: whole-number day 1–30 inclusive; room A or B. If any known value fails, label revise. Otherwise, if a required field is blank or unreadable, label review. Otherwise label ready. Work independently before comparing; this is an explicit-rule paper exercise, not a trained model.

The six practice cards

Card	Day	Room
P	1	A
Q	30	B
R	31	A
S	18	blank
T	31	unreadable
U	unreadable	B

Label, compare and investigate

1. Each partner records all six labels and a reason without seeing the other's answers. Compare corresponding cards and calculate matching labels / 6; keep the fraction even if all agree. For each disagreement, inspect the printed fields and guide before choosing a correction. Agreement alone cannot identify correct labels. If the guide is clear and one person misapplied it, correct the application rather than changing the guide. Preserve original answers, corrected labels, reasons and version; an intentional policy change gets a new version.

Keep the evidence record

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** fresh transfer
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

What did the hardest test teach you that the normal test did not?

> **Write or draw your reasoning and evidence.**

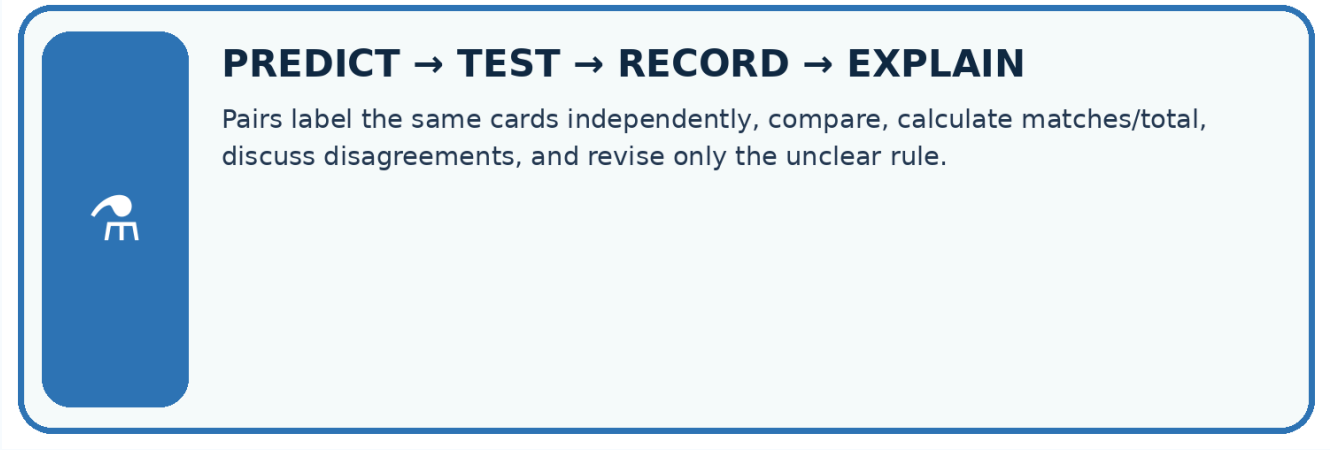
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Pairs **label** the supplied cards independently, compare matches/total and investigate **disagreement**s. Repair an incorrect application when the guide is clear; change an unclear or unsuitable guide explicitly with a new version and preserve the old record.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: apply the agreed 120-page policy

> Keep this new boundary case separate from supported 80-page practice.

The independent rules and records

Two **label**lers use book-length rules: fewer than 120 pages and at most 120 pages. **Label** books with 110,120,121 pages. The agreed policy is at most 120. Repair the **disagreement**.

Your first attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback after the attempt

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Changed independent case

Week 8 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Use a clear **label** rule: short means 80 pages or fewer. Two judges **label** an 80-page book differently. What shared instruction resolves this **disagreement**?

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Define ‘good match’ before **training** or evaluating the recommender.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Transfer practice from the original programme

Two **label**ers use 'short' to mean pages<100 and pages<=100. They disagree only on 100-page books. Write a shared definition and explain how to repair the existing **label**s.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

MAIN LESSON · SECTION 11

Define the recommender’s labels before evaluating them

> A task label does not describe a reader’s whole needs or abilities.

The checkpoint and metadata evidence

THIS WEEK’S CHECKPOINT Define ‘good match’ before **training** or evaluating the recommender.

Give Transparent Book Recommender metadata **label**s explicit definitions and boundary examples for reviewers.

Keep the system and human choice visible

SYSTEM MAP Reader need → allowed **input**s → transparent rule or score → ranked candidates → explanation → test record → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

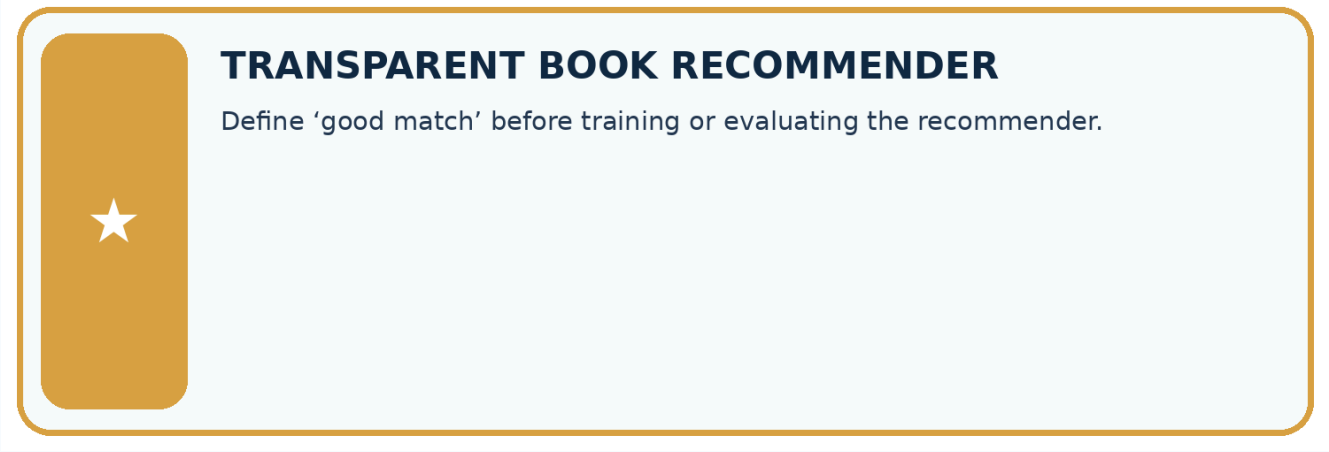
RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

▼ Earlier wording, alternatives and picture transcripts

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



Evidence for your project

MAIN LESSON · SECTION 16

Finish: explain the criterion, not just the label

> Recall an equality case and a limit of agreement.

Your unaided explanation

- Close your notes. Explain this goal in your own words: Apply a **category** definition using explicit **evidence** and boundary rules

My deciding step, evidence and one remaining question

Return on a changed case

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Use the words to explain a particular step, record or limit.

Core vocabulary

Word	My own example	Not this / boundary
label	The recorded target category or value attached to an example.	
operational definition	A precise rule explaining how a concept will be observed, measured or labelled in this activity.	
agreement	How often two judgments match; matching judgments can still both be wrong.	
ambiguous	Having more than one reasonable interpretation given the available information.	

Terms for the connected explanation

Word	Meaning
Category	A group defined by stated membership criteria

Word	Meaning
Annotator	A person who assigns labels to examples
Disagreement	A case where annotators assign different labels to the same example

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<summary>Earlier wording, alternatives and picture transcripts</summary>

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.



</details>


2
OPERATIONAL DEFIN
a testable rule for applying
a label


3
AGREEMENT
how often labelers match



Words used in the closer explanation

▼ **Choose a stronger thinking challenge**

REFERENCE · SECTION 10

Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

Pick your next step

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

- My strongest **evidence** of independent understanding:

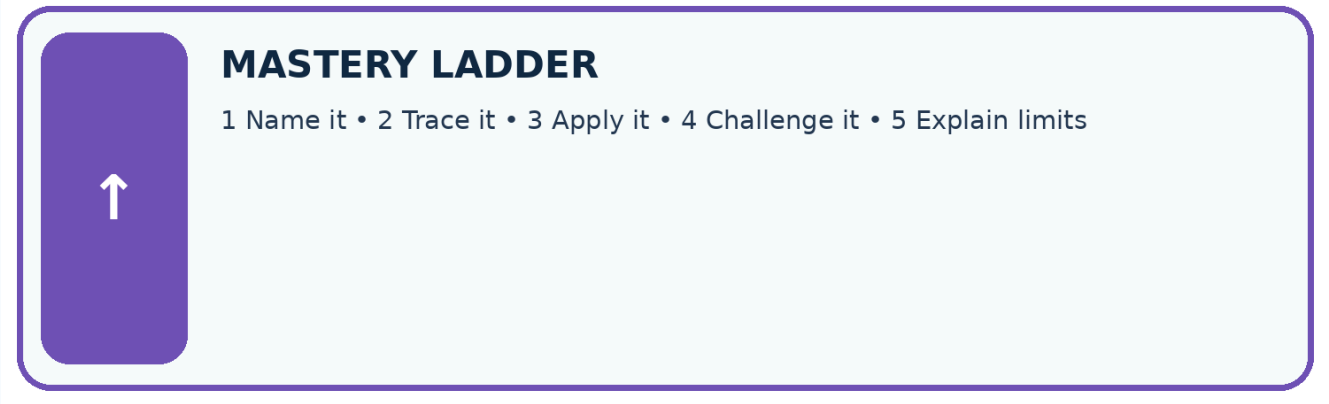
Write or draw your reasoning and evidence.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



▼ Reference: check your explanation

Reference: check your explanation

> Use after practice. Changed independent case is the selected independent case.

Check the evidence and its limit

- 1. My fresh example, trace, calculation, or design:
- 2. My **evidence** that it works (include a boundary or missing-data case):
- 3. One limitation, fairness concern, or situation needing human review:

> **Write or draw your reasoning and evidence.**

> **Write or draw your reasoning and evidence.**

Self-check and record

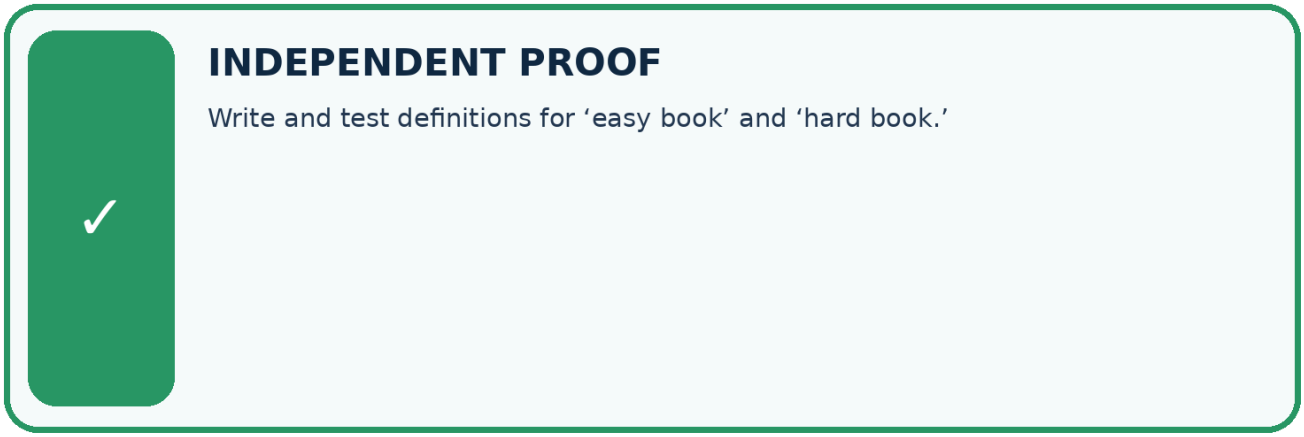
SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

▼ **Earlier wording, alternatives and picture transcripts**

INDEPENDENT MASTERY PROOF

Checklist for explaining this week's idea

Use after practice. Changed independent case is the selected independent case.



EXPLANATION CHECK Use a stated operational definition, retain the equality case and distinguish agreement from correctness.

Write or draw your reasoning and evidence.

▼ **Extra practice: Define Short before labelling books**

REFERENCE · SECTION 13

Extra practice: Define Short before labelling books

> Try the question before opening its worked explanation.

Read the supplied case and question

Short means fewer than 100 pages. Two **annotator**s disagree about a 100-page book.

Resolve the **label**.

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

It is not Short under the stated definition. Agree operational boundaries before **label**ling.

Evidence or stage	Value or result
99 pages	Short
100 pages	Not Short
101 pages	Not Short

▼ Earlier wording, alternatives and picture transcripts

Week 8 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Define Short before labelling books

Short means fewer than 100 pages. Two **annotator**s disagree about a 100-page book.

Resolve the **label**.

It is not Short under the stated definition. Agree operational boundaries before **label**ling.

▼ **Reference: original lesson checklist**

REFERENCE · SECTION 15

Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

Targets and evidence

By the end, I can	How I will show it
Apply a category definition using explicit evidence and boundary rules	Explain the deciding step, show a trace or test, and state one limit.
Distinguish a known failure from information that is missing or unreadable	Explain the deciding step, show a trace or test, and state one limit.
Use disagreement to improve a labelling guide without mistaking agreement for truth	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: Two fictional catalogue reviewers **label** books compact or extended.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 17

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- Two fictional catalogue reviewers **label** books compact or extended.
- Which **label**led step matches this detail? A must use <=80; retain the **disagreement** and changed policy implementation.
- 1. Repair boundary
- 2. Reviewer B
- 3. Reviewer A
- 4. Shared definition

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> ---
>
> ---
>
> ---
>
> ---

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Repair boundary A must use <=80; retain the **disagreement** and changed policy implementation.

The reviewers agree on two of three examples, but that fraction does not decide which definition to keep. The documented policy decides the equality case. Repairing A's implementation aligns its **label**s with the agreed meaning.

Rechecking 79 and 81 ensures the boundary fix preserves adjacent decisions.

Changed-case question

- Changed-case practice: Change the documented policy to fewer than 80 instead. Which conclusion is supported by the supplied facts?
- 1. A's original **label**s now match; B must change its equality behavior.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: A's original **label**s now match; B must change its equality behavior. The authoritative definition changed, not the physical books.

Repair and save your work

Review Rows Columns and Data Types (ai-2/7). Goal: Specify a table schema. Short practice: Which two rows originally fail, and do they fail for the same reason?. Return to this week and explain what you changed.

Review Loops and Stopping (ai-2/5). Goal: Trace variable values before and after each loop iteration. Short practice: Why is report 0 correct for the original loop, and why is equality-only stopping risky here?.

Return to this week and explain what you changed.

Review Decompose a Goal (ai-2/2). Goal: Distinguish a function definition from a function call. Short practice: What happens visibly if the caller stores needed but omits print(needed)?.

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

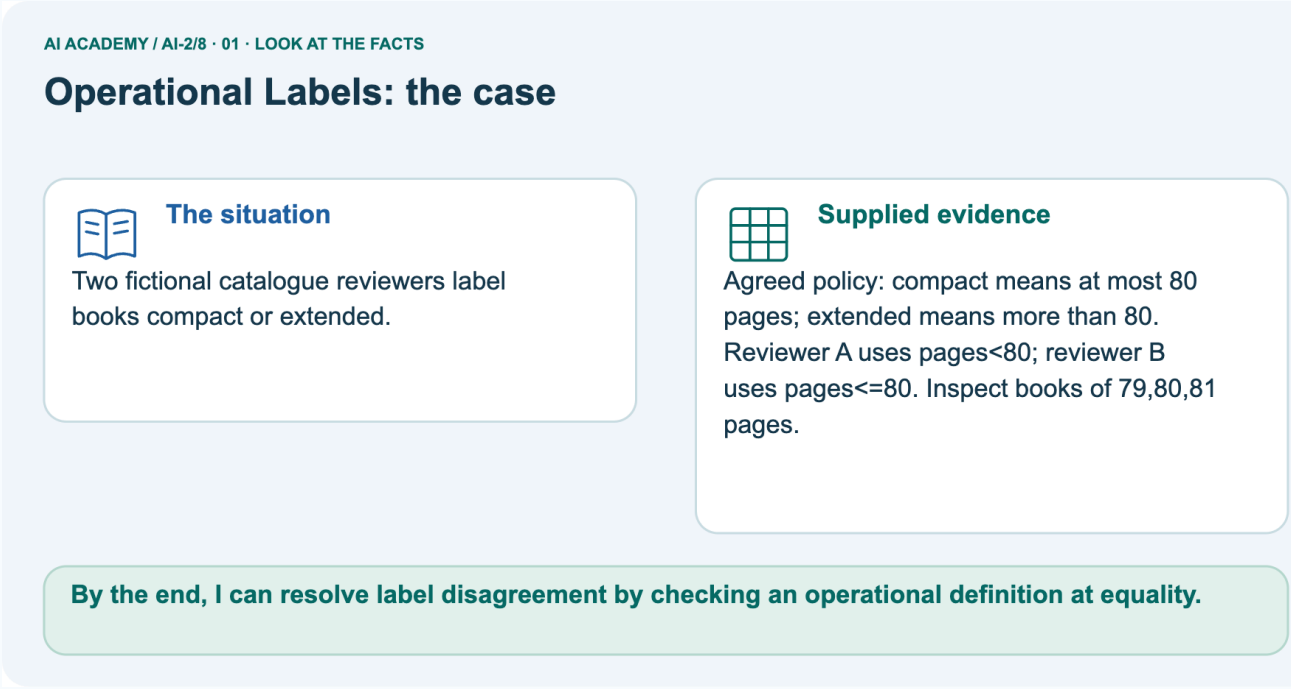
▼ **Reference: the six-picture journey**

REFERENCE · SECTION 18

Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

01. Operational Labels: the case



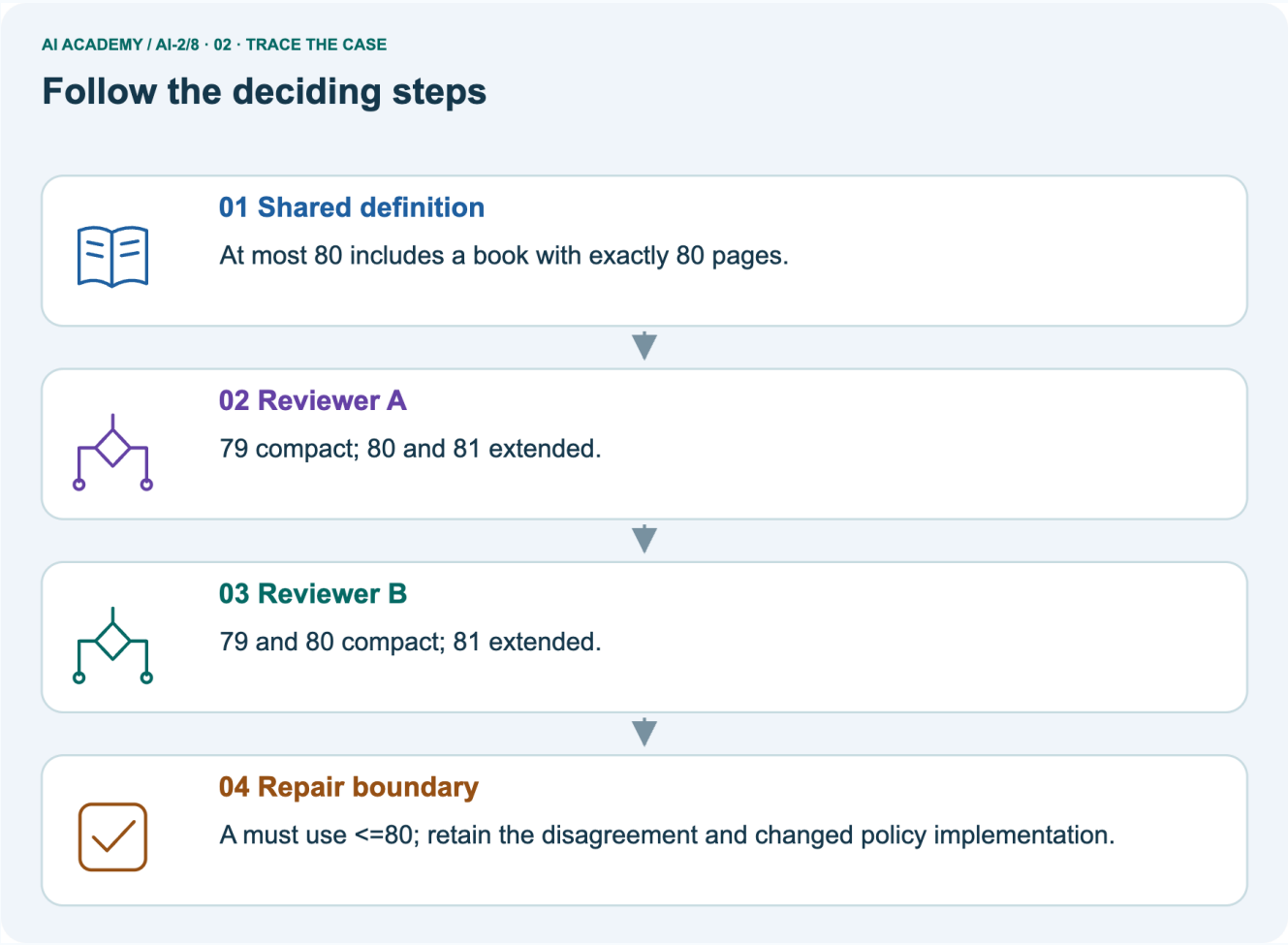
Follow the labelled picture with the matching explanation. Open the vector diagram

Two fictional catalogue reviewers **label** books compact or extended. Supplied facts: Agreed policy: compact means at most 80 pages; extended means more than 80. Reviewer A uses pages<80; reviewer B uses pages<=80.

Inspect books of 79,80,81 pages.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

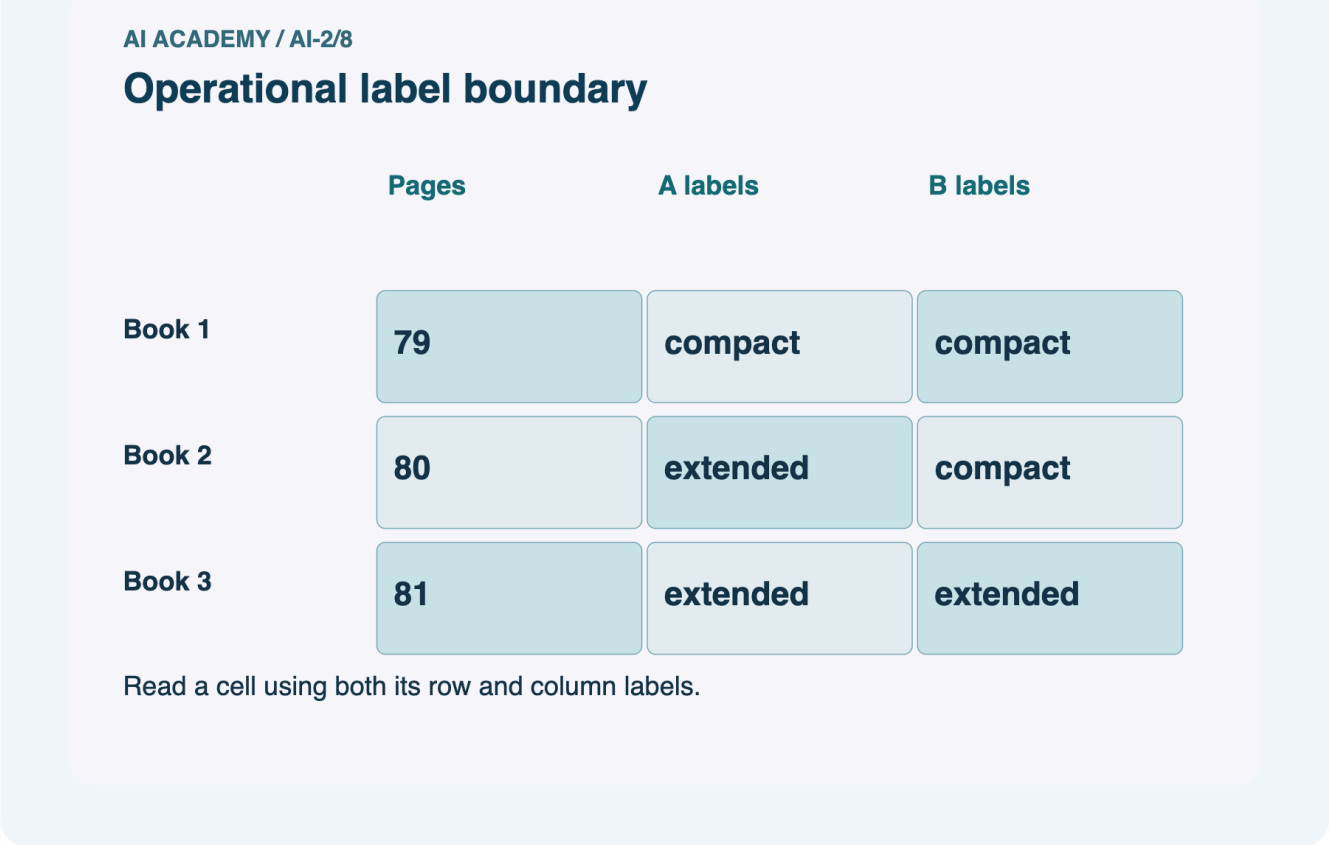


Follow the labelled picture with the matching explanation. Open the vector diagram

Shared definition: At most 80 includes a book with exactly 80 pages. → Reviewer A: 79 compact; 80 and 81 extended. → Reviewer B: 79 and 80 compact; 81 extended. → Repair boundary: A must use <=80; retain the **disagreement** and changed policy implementation.

Point to each numbered stage and explain which recorded **input** it uses.

03. Operational label boundary

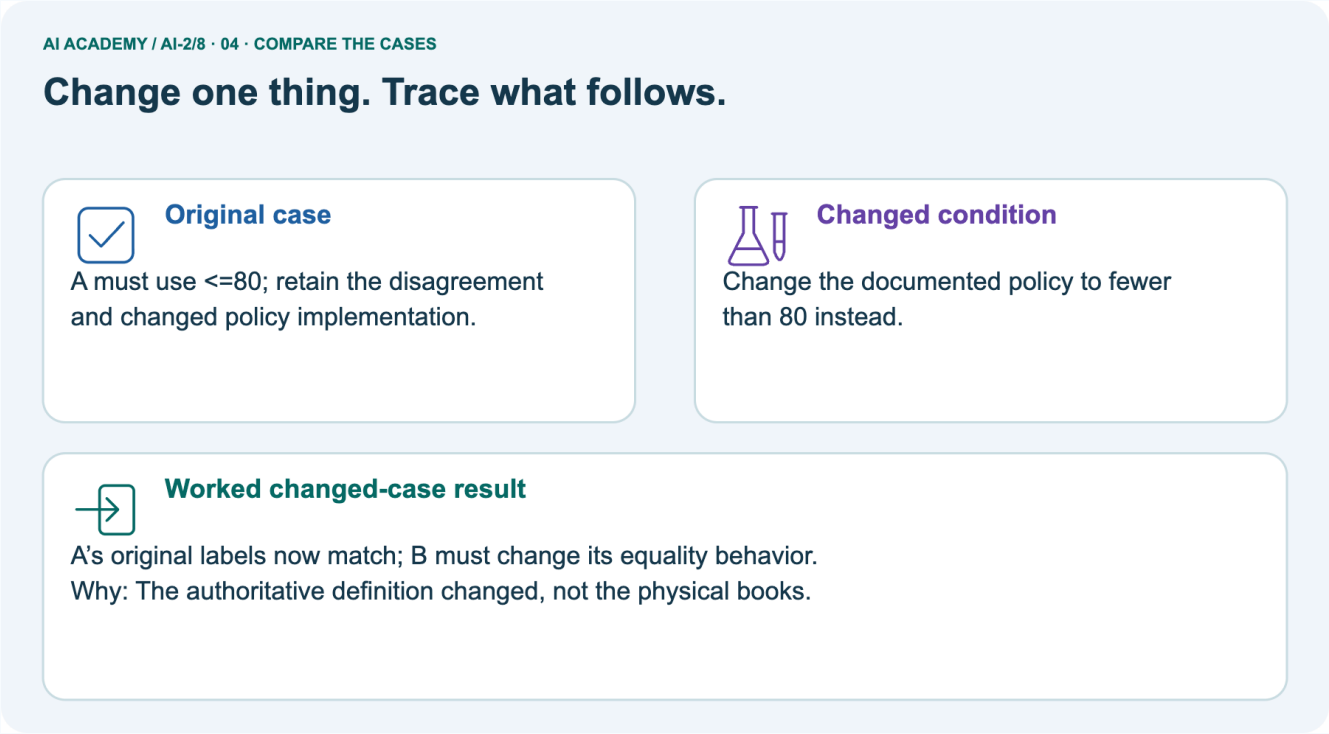


Follow the labelled picture with the matching explanation. Open the vector diagram

Operational **label** boundary. Book 1: Pages = 79, A **label**s = compact, B **label**s = compact. Book 2: Pages = 80, A **label**s = extended, B **label**s = compact. Book 3: Pages = 81, A **label**s = extended, B **label**s = extended.

Read the **label**s and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



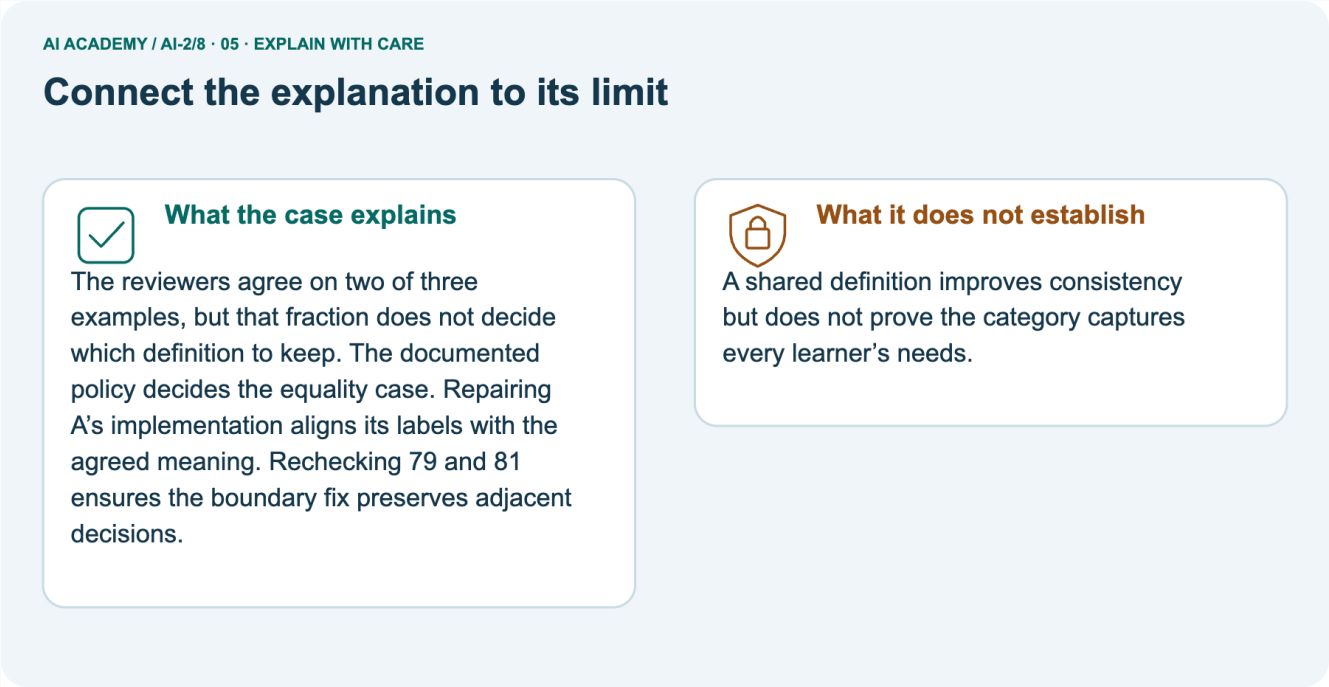
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: A must use <=80; retain the **disagreement** and changed policy implementation. Changed condition: Change the documented policy to fewer than 80 instead.

Expected result: A's original **label**s now match; B must change its equality behavior. Reason: The authoritative definition changed, not the physical books.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



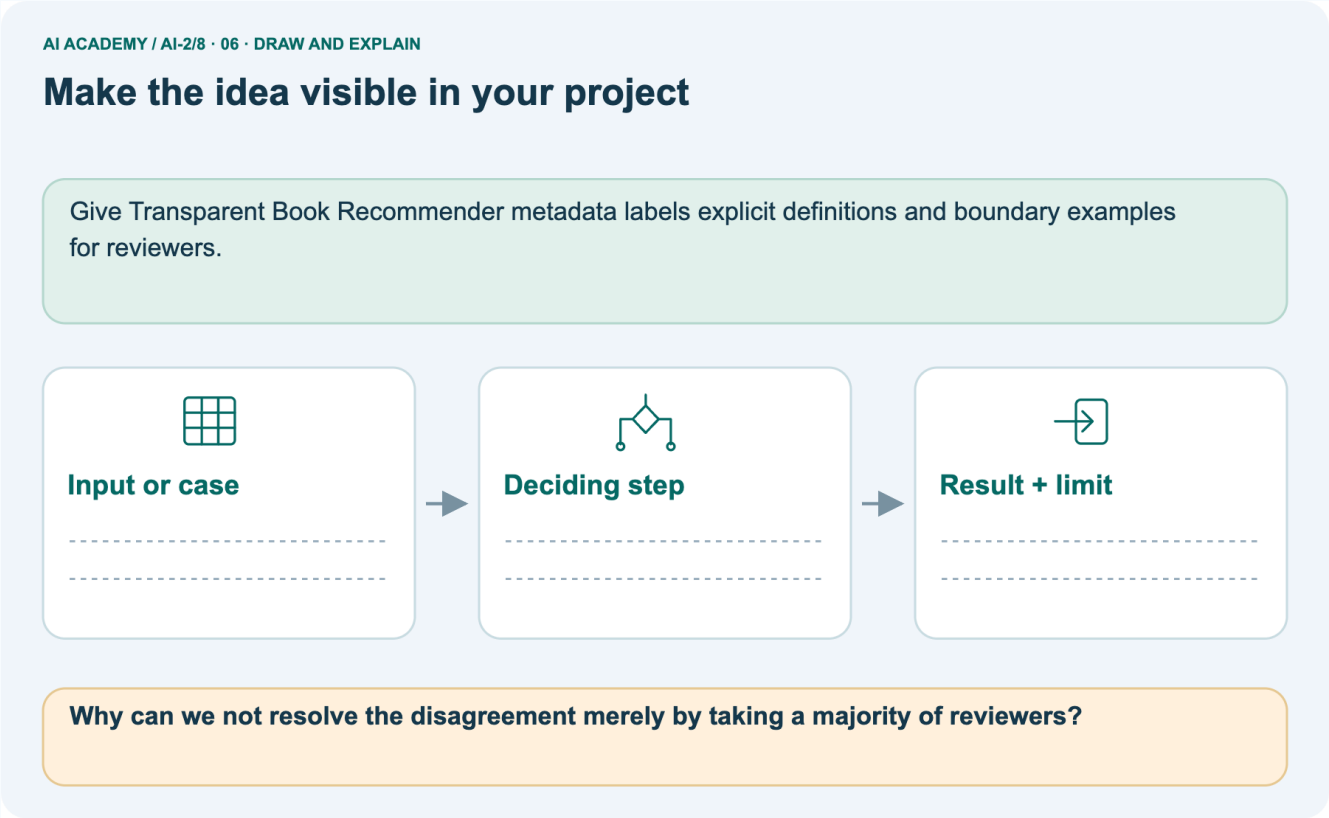
Follow the labelled picture with the matching explanation. Open the vector diagram

The reviewers agree on two of three examples, but that fraction does not decide which definition to keep. The documented policy decides the equality case. Repairing A's implementation aligns its **label**s with the agreed meaning.

Rechecking 79 and 81 ensures the boundary fix preserves adjacent decisions. Limit: A shared definition improves consistency but does not prove the **category** captures every learner's needs.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Give Transparent Book Recommender metadata **label**s explicit definitions and boundary examples for reviewers. Draw your **input**, deciding step, result and limit.

Supported question: Why can we not resolve the **disagreement** merely by taking a majority of reviewers?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Operational Labels: the case

02. Follow the deciding steps

03. Operational label boundary

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.